

Practical Algorithms and Fixed Parameter Tractability for the Single Individual SNP Haplotyping Problem

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Abstract

Single nucleotide polymorphisms (SNPs) are the most frequent form of human genetic variation, of foremost importance for a variety of applications including medical diagnostic, phylogenies and drug design. The complete SNPs sequence information from each of the two copies of a given chromosome in a diploid genome is called a haplotype. The Haplotypeing Problem for a single individual is as follows: Given a set of fragments from one individual's DNS, find a maximally consistent pair of SNPs haplotypes (one per chromosome copy) by removing data errors related to sequencing errors, repeats, and paralogous recruitment. Two versions of the problem, i.e. the Minimum Fragment Removal (MFR) and the Minimum SNP Removal (MSR), are considered.

Polymorphisms

A polymorphism is a feature

- common to every
- not identical in everybody
- the possible variants *alleles* are just a few
- e.g. eye-color or blood-type

Example: (eye-color)

Black eye	G	A	T	A	T	T	C	G	T	A	C	G	G	A	-	T
Brown eye	G	A	T	G	T	T	C	G	T	A	C	T	G	A	A	T
Blue eye	G	A	T	A	T	T	C	G	T	A	C	G	G	A	A	T

Single Nucleotide Polymorphism

- SNP arises from mutation.
- Mutation nucleotides become SNPs when observed frequency $> 1\%$ in a population.
 - SNP: DNA single base variations found $> 1\%$
 - Mutation: DNA single base variation found $< 1\%$

General Population	A	C	T	T	A	G	C	T	T	94.0%
SNP	A	C	T	T	A	G	C	T	C	6.0%

General Population	A	C	T	T	A	G	C	T	T	99.9%
Mutation	A	C	T	T	A	G	C	T	C	0.1%

Single Nucleotide Polymorphism (Contd.)

- All human share 99.9% the same DNA sequence.
 - SNPs occur about 600 base pairs.
 - 90% of human genome variation comes SNPs.
 - The human genome contains about 3 million SNPs.
- Because of the A-T/C-G complement, a SNP can have only two variants: (AT) or (CG).
- A SNP is variable with two states:
 - Major allele: Allele (i.e., (AT) or (CG)) $> 50\%$.
 - Minor allele: Allele $< 50\%$

Haplotype

- A set of closely linked SNPs located on one chromosome, which tend to be inherited together (not easily separated by recombination).

Phenotype				SNP1								SNP2				SNP3	T
Black eye	G	A	T	A	T	T	C	G	T	A	C	G	G	A	-	T	
Brown eye	G	A	T	G	T	T	C	G	T	A	C	T	G	A	A	T	
Blue eye	G	A	T	A	T	T	C	G	T	A	C	G	G	A	A	T	

Haplotypes			
A	G	-	2/6
G	T	A	3/6
A	G	A	1/6

The Haplotyping problem

- Single Individual: Given genomic data of one individual, determine 2 haplotypes (one per chromosome).
- Population: Given genomic data of k individuals, determine (at most) $2k$ haplotypes (one per chromosome/individuals).
- For the individual problem, input is erroneous haplotype data, from sequencing.
- For the population problem, data is ambiguous genotype data, from screening.

The Single-Individual Haplotyping Problem

- Given a set of fragments obtained by DNS sequencing from the two copies of a chromosome, find the smallest number of errors so that there exist two haplotype compatible with all the (corrected) fragments observed.
- Bad fragments: Find the minimum number of fragments to ignore.
- Bad SNPs: Find the minimum number of SNPs to ignore.

SNP Matrix

- A SNP matrix M :

	1	2	3	4	5	6
1	A	B	-	A	A	B
2	B	A	-	-	B	-
3	-	A	B	A	B	A
4	-	A	B	-	B	A
5	B	-	A	B	A	-

- A SNP matrix M is called *error-free* if we can partition the rows (fragments) into two classes of non-conflicting fragments.

Conflict

- For a SNP s , two fragments f and g are said to conflict on s if $M[f, s] = A$ and $M[g, s] = B$ or vice-versa.
- f and g are said to *conflict* if exists a SNP s such that they conflict on s , otherwise are said to *agree*.
- Two SNPs s and t are said to be in conflict if there exists fragments f and g such that $M[f, a], M[g, a], M[f, c], M[g, c] \neq -$ and the boolean value $(M[f, a] = M[g, a])$ is the negation of $(M[f, c] = M[g, c])$.

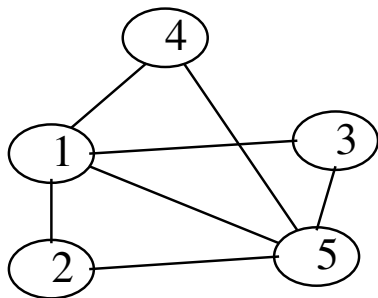
Conflict Graph

A SNP matrix M :

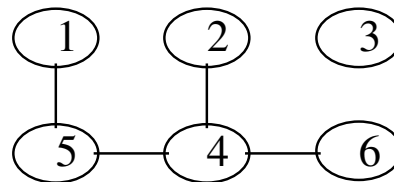
	1	2	3	4	5	6
1	A	B	-	A	A	B
2	B	A	-	-	B	-
3	-	A	B	A	B	A
4	-	A	B	-	B	A
5	B	-	A	B	A	-

(a). The Fragment conflict graph $G_F(M)$.

(b). The SNP conflict graph $G_S(M)$.



(a)



(b)

M is *error-free* if and only if $G_F(M)$ is *bipartite*.

Gapless Matrix

- A gapless fragment i is one for which the As and Bs appear consecutively, with no -s between holes between two non-hole symbols.
- Example:
gapless : -ABABBBBA---
two gaps : -AB---B-AB-
- Gapless Matrix: All fragments are gapless.

MSR and MFR

- MSR Minimum SNP Removal - Find a minimum number of columns (SNPs) whose removal makes M error-free.
- MFR Minimum Fragment Removal - Find a minimum number of rows (fragments) whose removal makes M error-free.

MSR (Gapless Case)

- Notation:

M : A $m \times n$ SNP matrix.

$K[j]$: For $j \leq n$, the maximum number of columns that can be kept to make M error-free.

- The solution to the problem is given by

$$\max_{j \in \{0, \dots, n\}} K[j]$$

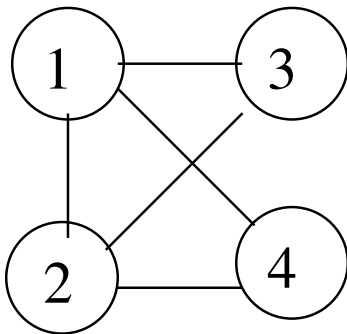
- For every j , we define $OK(j)$ as the set of those i with $i < j$ such that clounms i and j do not conflict.
For every j

$$K[j] = 1 + \max_{i \in OK(j)} K[i]$$

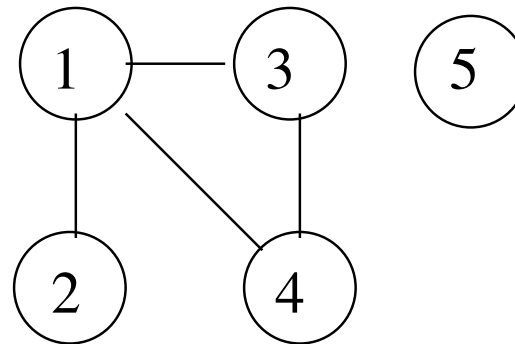
Example

A SNP matrix M :

	1	2	3	4	5
1	-	-	A	B	-
2	A	B	A	A	-
3	-	-	B	B	-
4	A	A	B	B	A

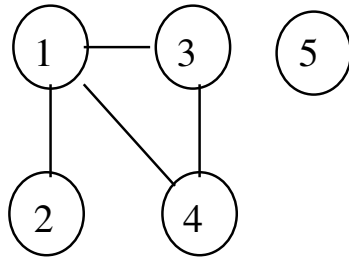


The fragment conflict graph



The SNP conflict graph

Example (Contd.)



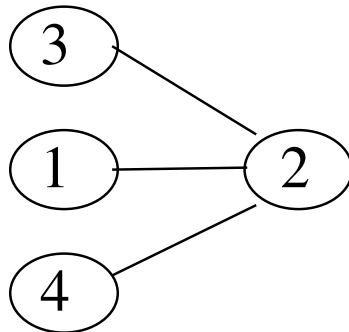
The SNP conflict graph

- $K[0] = 0$
- $j = 1, \text{OK}(1) = \{0\} \Rightarrow K[1] = 1$
- $j = 2, \text{OK}(2) = \{0\} \Rightarrow K[2] = 1$
- $j = 3, \text{OK}(3) = \{0, \underline{2}\} \Rightarrow K[3] = 2$
- $j = 4, \text{OK}(4) = \{0, \underline{2}\} \Rightarrow K[4] = 2$
- $j = 5, \text{OK}(5) = \{0, 1, 2, \underline{3}, \underline{4}\} \Rightarrow K[5] = 3$

Example (Contd.)

The matrix M' is obtained from M by dropping $n - k$ columns.

	2	4	5
1	-	B	-
2	B	A	-
3	-	B	-
4	A	B	A



The fragment conflict graph

$G_F(M)$ is *bipartite*. $\Rightarrow M'$ is *error-free*.

The Time Complexity

- Determining if two SNPs are in conflict can be done in time $O(m)$.
- the cost of creating the $OK(j)$ is $O(mn^2)$.
This dominates the cost $O(n^2)$ of solving all equations.
- $\Rightarrow O(mn^2)$

Next

- MFR
- Dealing with Gaps
- The Population Haplotyping Problem

Thanks

Thanks